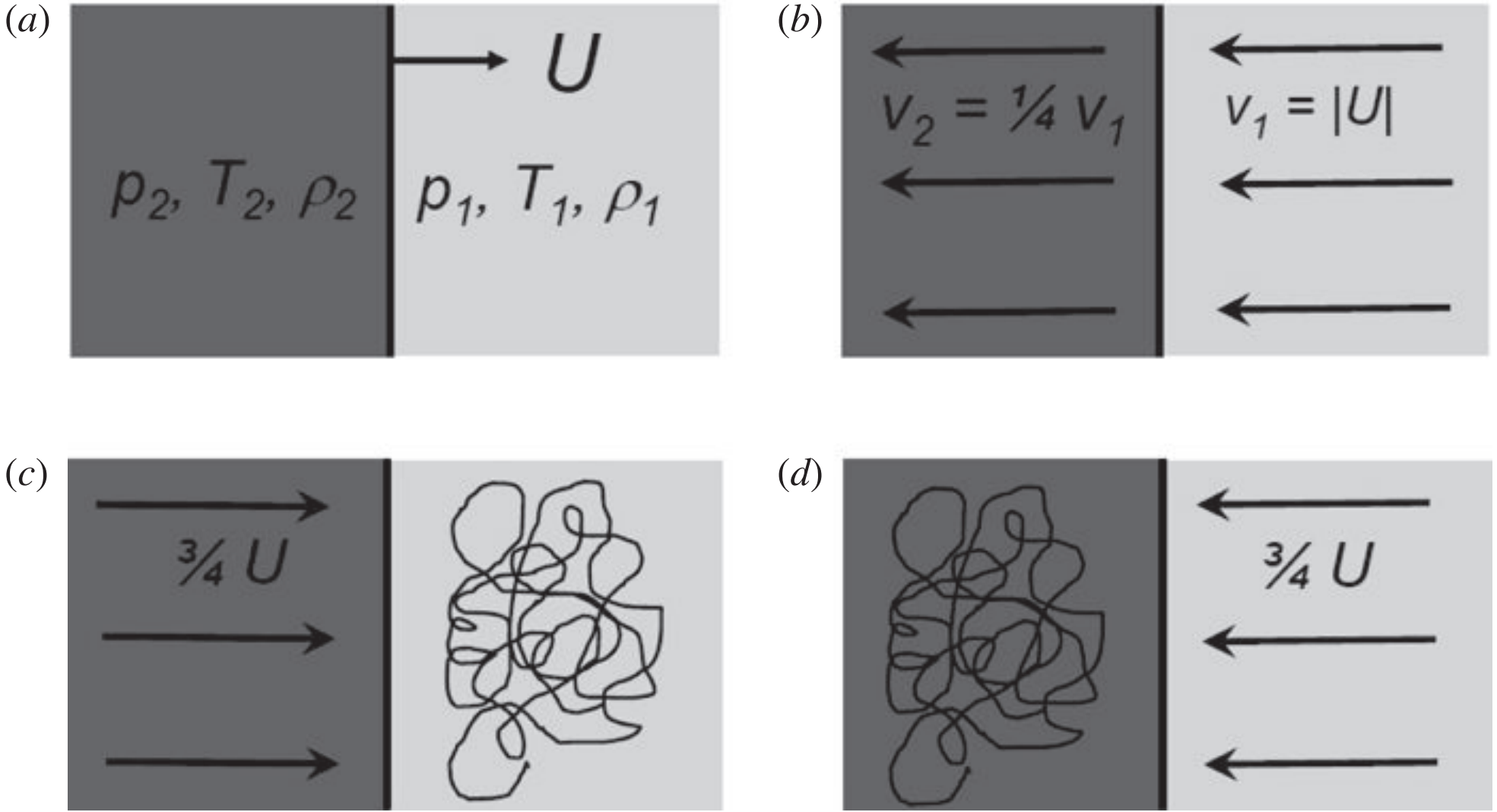

(a)
U
p2, T2, ρ2
p1, T1, ρ1
(b)
v2 = ¼ v1
v1 = |U|
(c)
¾ U
(d)
¾ U